

**LAPORAN PRAKTIKUM
APLIKASI BERBASIS PLATFORM**

MODUL 7&8



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**PROGRAM STUDI S1 INFORMATIKA
FAKULTAS INFORMATIKA
TELKOM UNIVERSITY PURWOKERTO
2025/2026**

Tugas

Source Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>COTS 1 - Data Table</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdn.datatables.net/1.13.4/css/dataTables.bootstrap5.min.css"
rel="stylesheet">
</head>
<body class="bg-light">

  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="#">Sistem Akademik</a>
      <div class="navbar-nav">
        <a class="nav-link active" href="index.html">Daftar Mahasiswa (Table)</a>
        <a class="nav-link" href="form.html">Tambah Mahasiswa (Form)</a>
        <a class="nav-link" href="crud.html">Log Aktivitas (CRUD)</a>
      </div>
    </div>
  </nav>

  <div class="container my-5">
    <div class="card shadow-sm">
      <div class="card-header bg-primary text-white d-flex justify-content-between
align-items-center">
        <h5 class="mb-0">Daftar Mahasiswa Aktif</h5>
        <a href="form.html" class="btn btn-light btn-sm fw-bold">+ Tambah Data</a>
      </div>
      <div class="card-body">
        <table id="studentTable" class="table table-striped table-bordered w-100">
          <thead>
            <tr>
              <th>ID</th>
              <th>NIM</th>
              <th>Nama Lengkap</th>
              <th>Program Studi</th>
              <th>Email</th>
              <th>Aksi</th>
            </tr>
          </thead>
          <tbody></tbody>
        </table>
      </div>
    </div>
  </div>

  <script src="https://code.jquery.com/jquery-3.6.4.min.js"></script>
```

```

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
<script src="https://cdn.datatables.net/1.13.4/js/jquery.dataTables.min.js"></script>
<script
src="https://cdn.datatables.net/1.13.4/js/dataTables.bootstrap5.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/sweetalert2@11"></script>

<script>
    $(document).ready(function() {
        // Menggunakan data JSON untuk DataTables via AJAX (Sesuai Poin 5)
        const table = $('#studentTable').DataTable({
            "ajax": "/api/students",
            "columns": [
                { "data": "id" },
                { "data": "nim" },
                { "data": "nama" },
                { "data": "prodi" },
                { "data": "email" },
                {
                    "data": "id",
                    "render": function(data) {
                        return `<button class="btn btn-danger btn-sm btn-delete" data-
id="${data}">Hapus</button>`;
                    }
                }
            ]
        });

        // Fitur Hapus CRUD berbasis AJAX dan SweetAlert (Sesuai Poin 4 & 5)
        $('#studentTable').on('click', '.btn-delete', function() {
            const id = $(this).data('id');
            Swal.fire({
                title: 'Apakah Anda yakin?',
                text: "Data yang dihapus tidak bisa dikembalikan!",
                icon: 'warning',
                showCancelButton: true,
                confirmButtonColor: '#d33',
                cancelButtonColor: '#3085d6',
                confirmButtonText: 'Ya, hapus!'
            }).then((result) => {
                if (result.isConfirmed) {
                    $.ajax({
                        url: `/api/students/${id}`,
                        type: 'DELETE',
                        success: function(response) {
                            Swal.fire('Terhapus!', response.message, 'success');
                            table.ajax.reload(); // Reload data JSON table otomatis
                        }
                    });
                }
            });
        });
    });
</script>
</body>
</html>

```

Form.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>COTS 1 - Form Input</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light">

  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="#">Sistem Akademik</a>
      <div class="navbar-nav">
        <a class="nav-link" href="index.html">Daftar Mahasiswa (Table)</a>
        <a class="nav-link active" href="form.html">Tambah Mahasiswa (Form)</a>
        <a class="nav-link" href="crud.html">Log Aktivitas (CRUD)</a>
      </div>
    </div>
  </nav>

  <div class="container my-5" style="max-width: 600px;">
    <div class="card shadow-sm">
      <div class="card-header bg-success text-white">
        <h5 class="mb-0">Form Tambah Mahasiswa</h5>
      </div>
      <div class="card-body">
        <form id="studentForm" class="needs-validation" novalidate>
          <div class="mb-3">
            <label for="nim" class="form-label">NIM</label>
            <input type="text" class="form-control" id="nim"
required placeholder="Contoh: A11.2022.12345">
          </div>
          <div class="mb-3">
            <label for="nama" class="form-label">Nama Lengkap</label>
            <input type="text" class="form-control" id="nama" required>
          </div>
          <div class="mb-3">
            <label for="prodi" class="form-label">Program Studi</label>
            <select class="form-select" id="prodi" required>
              <option value="">Pilih Program Studi...</option>
              <option value="D3 Keperawatan">D3 Keperawatan</option>
              <option value="Teknik Informatika">Teknik Informatika</option>
              <option value="Sistem Informasi">Sistem Informasi</option>
            </select>
          </div>
          <div class="mb-3">
            <label for="email" class="form-label">Email</label>
            <input type="email" class="form-control" id="email" required
placeholder="nama@mhs.id">
          </div>
          <button type="submit" class="btn btn-success w-100">Simpan
Data</button>
        </form>
      </div>
    </div>
  </div>
```

```

    </div>
</div>

<script src="https://code.jquery.com/jquery-3.6.4.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/sweetalert2@11"></script>
<script>
    $('#studentForm').on('submit', function(e) {
        e.preventDefault();

        // Validasi Bootstrap
        if (!this.checkValidity()) {
            e.stopPropagation();
            $(this).addClass('was-validated');
            return;
        }

        const dataMahasiswa = {
            nim: $('#nim').val(),
            nama: $('#nama').val(),
            prodi: $('#prodi').val(),
            email: $('#email').val()
        };

        // Post data ke API Express dalam format JSON
        $.ajax({
            url: '/api/students',
            type: 'POST',
            contentType: 'application/json',
            data: JSON.stringify(dataMahasiswa),
            success: function(response) {
                Swal.fire('Berhasil!', response.message, 'success').then(() => {
                    window.location.href = 'index.html'; // Redirect ke halaman tabel
                });
            }
        });
    });
</script>
</body>
</html>

```

crud.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>COTS 1 - CRUD Log</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
    <link href="https://cdn.datatables.net/1.13.4/css/dataTables.bootstrap5.min.css"
rel="stylesheet">
</head>
<body class="bg-light">

```

```

<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
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      <a class="nav-link" href="form.html">Tambah Mahasiswa (Form)</a>
      <a class="nav-link active" href="crud.html">Log Aktivitas (CRUD)</a>
    </div>
  </div>
</nav>

<div class="container my-5">
  <div class="card shadow-sm">
    <div class="card-header bg-dark text-white">
      <h5 class="mb-0">Histori Aktivitas Aplikasi (CRUD Log)</h5>
    </div>
    <div class="card-body">
      <table id="logTable" class="table table-hover table-bordered w-100">
        <thead class="table-secondary">
          <tr>
            <th>ID Log</th>
            <th>Deskripsi Aktivitas</th>
            <th>Waktu Eksekusi</th>
          </tr>
        </thead>
        <tbody></tbody>
      </table>
    </div>
  </div>
</div>

<script src="https://code.jquery.com/jquery-3.6.4.min.js"></script>
<script src="https://cdn.datatables.net/1.13.4/js/jquery.dataTables.min.js"></script>
<script
src="https://cdn.datatables.net/1.13.4/js/dataTables.bootstrap5.min.js"></script>
<script>
  $(document).ready(function() {
    $('#logTable').DataTable({
      "ajax": "/api/logs",
      "columns": [
        { "data": "id" },
        { "data": "aktivitas" },
        { "data": "waktu" }
      ],
      "order": [[0, "desc"]] // Tampilkan log terbaru di atas
    });
  });
</script>
</body>
</html>

```

server.js

```

const express = require('express');
const path = require('path');
const app = express();

```

```

const PORT = 3000;

// Middleware untuk parsing data
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname, 'public')));

// Mock Data Awal (Format JSON)
let students = [
  { id: 1, nim: 'A11.2022.12345', nama: 'Anggita Prishelia', prodi: 'D3 Keperawatan',
    email: 'anggita@mhs.id' },
];

let logs = [
  { id: 1, aktivitas: 'Membaca seluruh data mahasiswa', waktu: new
Date().toLocaleString() }
];

// ===== API ENDPOINTS (JSON DATA) =====

// Ambil semua data mahasiswa
app.get('/api/students', (req, res) => {
  res.json({ data: students });
});

// Tambah data mahasiswa
app.post('/api/students', (req, res) => {
  const { nim, nama, prodi, email } = req.body;
  const newStudent = { id: students.length + 1, nim, nama, prodi, email };
  students.push(newStudent);

  logs.push({ id: logs.length + 1, aktivitas: `Menambahkan mahasiswa: ${nama}`, waktu:
new Date().toLocaleString() });
  res.status(201).json({ message: 'Data berhasil ditambahkan', data: newStudent });
});

// Hapus data mahasiswa
app.delete('/api/students/:id', (req, res) => {
  const { id } = req.params;
  const studentIndex = students.findIndex(s => s.id == id);
  if (studentIndex !== -1) {
    const deleted = students.splice(studentIndex, 1);
    logs.push({ id: logs.length + 1, aktivitas: `Menghapus mahasiswa:
${deleted[0].nama}`, waktu: new Date().toLocaleString() });
    res.json({ message: 'Data berhasil dihapus' });
  } else {
    res.status(404).json({ message: 'Data tidak ditemukan' });
  }
});

// Ambil data log aktivitas CRUD
app.get('/api/logs', (req, res) => {
  res.json({ data: logs });
});

// Jalankan Server
app.listen(PORT, () => {
  console.log(`Server COTS 1 berjalan di http://localhost:${PORT}`);
});

```

```
});
```

Output Program

Sistem Akademik					
Daftar Mahasiswa (Table) Tambah Mahasiswa (Form) Log Aktivitas (CRUD)					
Daftar Mahasiswa Aktif					
Show 10 entries				Search:	
ID	NIM	Nama Lengkap	Program Studi	Email	Aksi
1	2311102224	Rayvaldo Ryzantha Damaysa	Teknik Informatika	rayvaldord@gmail.com	Hapus
Showing 1 to 1 of 1 entries					Previous 1 Next

Deskripsi Program

Program ini merupakan sebuah pencatatan daftar mahasiswa untuk sistem akademik yang berisi nama lengkap, nim, prodi, dan email. Program berisi tiga halaman yaitu index.html, form.html, dan crud.html.

Sistem Akademik					
Daftar Mahasiswa (Table) Tambah Mahasiswa (Form) Log Aktivitas (CRUD)					
Daftar Mahasiswa Aktif					
Show 10 entries				Search:	
ID	NIM	Nama Lengkap	Program Studi	Email	Aksi
1	2311102224	Rayvaldo Ryzantha Damaysa	Teknik Informatika	rayvaldord@gmail.com	Hapus
Showing 1 to 1 of 1 entries					Previous 1 Next

Halaman utama yaitu index.html, berisi tabel daftar mahasiswa aktif, yang menunjukkan ID, NIM, Nama Lengkap, Program Studi, Email, dan kolom untuk menghapus data mahasiswa.

Sistem Akademik Daftar Mahasiswa (Table) **Tambah Mahasiswa (Form)** Log Aktivitas (CRUD)

Form Tambah Mahasiswa

NIM
Contoh: A11.2022.12345

Nama Lengkap

Program Studi
Pilih Program Studi...

Email
nama@mhs.id

Simpan Data

Halaman form.html adalah halaman untuk menambah data dari mahasiswa, yang bila disimpan, akan muncul di tabel daftar mahasiswa di index.html.

Sistem Akademik Daftar Mahasiswa (Table) Tambah Mahasiswa (Form) **Log Aktivitas (CRUD)**

Histori Aktivitas Aplikasi (CRUD Log)

Show 10 entries Search:

ID Log	Deskripsi Aktivitas	Waktu Eksekusi
10	Menambahkan mahasiswa: Rayvaldo Ryzantha Damaysa	16/06/2026, 16.05.37
9	Menghapus mahasiswa: Rayvaldo Ryzantha Damaysa	16/06/2026, 12.44.08
8	Menambahkan mahasiswa: Rayvaldo Ryzantha Damaysa	16/06/2026, 12.43.56
7	Menghapus mahasiswa: afwgrgerg	16/06/2026, 12.39.27

Halaman crud.html adalah halaman untuk melihat histori atau log riwayat dari aktivitas crud yang dilakukan seperti menambah data atau menghapus data.

Link presentasi:

https://drive.google.com/file/d/1aKqeF2jw2Z_mDUUs80OCTHgZUkK_5i8D/view?usp=drive_link